Scalability & Future Plan

Date	04 july 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the Credit Card Approval Prediction System can be scaled to handle larger applicant volumes, growing datasets, and additional features. It also captures the team's roadmap for enhancing and evolving the project beyond its current demo-scale implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Flask app runs as a single instance and loads one saved model, with no load balancing or auto-scaling built in.	Response times degrade and the app may fail under high concurrent traffic, e.g. during peak application periods.	High
2	The model is trained on a static, historical dataset with no live connection to core banking systems.	Predictions can drift over time as real applicant profiles and lending conditions change.	Medium
3	No authentication, role-based access, or audit logging is implemented in the current web application.	Sensitive financial and personal applicant data could be exposed to unauthorized users, and there is no compliance trail.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask server handling requests sequentially; suited only for demo-scale traffic.	Containerize the app with Docker and deploy behind a load balancer (e.g. IBM Cloud Kubernetes Service) so multiple instances can auto-scale with demand.
2	Data Storage	Applicant dataset stored as a local CSV file used for training and reference.	Move applicant and prediction data to a managed cloud database (e.g. IBM Db2 or cloud object storage) to support growing volumes and easier retraining pipelines.
3	Performance	Best model is loaded and queried per request with no caching or batching.	Introduce model caching, batch-inference endpoints, and periodic model refresh, along with lightweight model formats for faster load times.
4	Security	No authentication, encryption, or audit trail for applicant data currently.	Add API-key or OAuth-based authentication, encrypt PII at rest and in transit, and maintain audit logs to meet financial compliance requirements.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Batch/CSV upload for bulk applicant screening	1-2 weeks	Enables compliance officers to screen many applicants at once, improving efficiency.
Phase 2	Automated model retraining pipeline with new applicant data	3-4 weeks	Keeps the model accurate over time and reduces the risk of prediction drift.
Phase 3	Add explainability (e.g., SHAP values) to prediction results	5-6 weeks	Increases transparency and trust by showing analysts and customers why an application was approved or rejected.
Phase 4	Role-based authentication and audit logging for compliance	7-9 weeks	Improves security and provides an audit trail for regulatory compliance in the banking domain.